



Angle Judgment: Is the Whole the Sum of its Parts?

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This study concerns whether the discrimination of a geometric angle depends on the orientations of its bounding lines or on angle size. In Experiment 1, thresholds for angle discrimination were measured in three observers for angles ranging from 15 to 180 deg, oriented either vertically or obliquely. Angle discrimination thresholds were found to depend primarily on angle size for most of the range of angles (angle-dependent, or Weber's law regime). However, in a small region near 90 deg (orientation-dependent regime) angle discrimination depends on the orientations of the bounding lines. When our data in the angle-dependent regime were fitted with a power function, the exponents were close to or <0.5 , suggesting that a step-increment approach was used to calculate angle. In Experiment 2, orientation discrimination thresholds for lines corresponding to the bounding lines of the vertically and obliquely oriented 15, 90 and 165 deg angles were measured. Confirming previous studies, a strong meridional anisotropy in line orientation discrimination was found for all three observers. The orientation discrimination thresholds were then used to predict the discrimination thresholds for the corresponding angles based on a simple statistical model. The predicted angle discrimination thresholds were worse than those measured empirically except for the tilted 90 deg angles. This result indicates that angle discrimination thresholds are not limited by the same noise as orientation discrimination for most angles except for the tilted 90 deg angle, where the limiting factor may be the precision in determining the orientations of the bounding lines. In Experiment 3, we show that angle discrimination is quite robust to small amounts of orientation jitter, suggesting that angle judgments are made at a level beyond the early filter representation. Copyright © 1996 Elsevier Science Ltd.

Oblique effect Angle discrimination Orientation discrimination Weber's law Spatial relation

INTRODUCTION

What are the elements of perception and how are they combined to form an integrated percept of visual stimuli? These are fundamental questions for understanding visual perception. Historically, two extreme theories have been proposed: the structuralist theory that the perception of whole figures is nothing more than the concatenation of primitive perceptual units, and the Gestalt theory that the perception of whole figures is an indivisible entity whose properties are not determined completely from the properties of their components. Both theories have their own supporting evidence [see Treisman (1986); Uttal (1981) for reviews]. More recently, an eclectic theory of perceptual representation has been proposed (Biederman, 1987; Palmer, 1977; Ullman, 1989). This theory synthesizes the holistic and atomistic approaches, postulating numerous levels of representation in the form

of hierarchical networks. At each level in a hierarchy, structural units are defined both holistically as a set of global properties and elementally as an organized set of parts. These parts are the structural units at the next-lower level in the hierarchy. The object as a whole has certain global properties as well as a set of component parts with specific perceptual relationships between them.

The present study concerns the perceptual representation of a simple two-part figure—an angle. Intuitively, the perceptual representation of an angle includes the representation of two lines as sub-parts, each part has its own properties such as orientation and length. But it is also important to realize that some global attributes of angles, i.e. conjunction and area, are not attributes of the component lines. Furthermore, the spatial relations between the parts of an angle are not represented in the properties of its parts. We are especially interested in whether the coding of the size of an angle (a global or relational property) is related to the orientation of the individual lines of an angle. This hypothesis is not unreasonable since once the visual system has extracted the orientation of the two component lines, it can easily compute the size of an angle by subtracting one orientation from the other. An alternative hypothesis is

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that the angular size is represented as a single holistic “chunk”, not dependent on the properties of its component parts.

Typically, perceptual representation of visual images has been studied with cognitive approaches, i.e. parsing figures, subjective goodness ratings of parts within figures, etc, or a perturbation technique (Foster, 1982). We approached the question of perceptual representation of angular size by comparing the performance of angle discrimination and line orientation discrimination. It is well established that our ability to judge the orientation of a stimulus is poorer for obliquely oriented stimuli than for those that are vertical or horizontal (Appelle, 1972). This orientation discrimination anisotropy, or “oblique effect”, has been demonstrated for a wide variety of stimuli (e.g. lines, gratings, and edges) and with different paradigms (Heeley & Buchanan-Smith, 1992; Heeley & Timney, 1988; Orban *et al.*, 1984a,b; Regan & Price, 1986). The rationale of this experiment is that if angle size is represented by extracting and comparing the orientation of its two component lines, then we should be able to predict the performance of angle discrimination from the orientation discrimination of its two lines.* To be more specific, when the two bounding lines of an angle lie in or close to the horizontal and/or vertical meridian, we would expect angle discrimination to be good, whereas when the two lines lie in or close to an oblique meridian, angle discrimination threshold should be worse. Indeed, this hypothesis has been tested recently using three dots corresponding to the three endpoints of a line-angle (Snippe & Koenderink, 1994). Snippe and Koenderink reported that their angle discrimination thresholds varied as a function of base angle and stimulus orientation, but their variations can all be understood from the meridional anisotropy for the discrimination of the orientations that define the angle. Therefore, they concluded that observers measure geometric angle (formed by the three dots) by simply comparing the visual orientations that define the angle. However, they also realized that different results might be derived by using angle stimuli made of two lines rather than three dots. For angles made of dots, the virtual line segments may need to form first internally for angle judgment and this requirement may force observers to use a *multilocal* strategy that analyses the stimulus through its components. For line angles, however, there is both multilocal information at the line segments’ endpoints and wholistic information (shape). Consequently, observers may directly wholistically extract the relevant information, instead of analyzing the stimulus components. In this case, angle discrimination may depend on the size of the angle rather than on the line orientations.

In the present study, angles made of two lines rather

than dots were used as visual stimuli. We describe three experiments in this paper. In Experiment 1, angle discrimination thresholds were measured for a series of angles, with angle stimuli either oriented vertically or tilted. There are two cases that are particularly interesting. In one case, two stimuli have the same angle (iso-angle), but their bounding lines are in different meridians. In the other case, two stimuli have a different size of angle, but their bounding lines share the same orientation (iso-orientation). These interesting conditions allow us to ask whether angle discrimination depends on the orientations of the bounding lines, or depends on the size of angle, or both. The secondary aim of Experiment 1 is to examine how angle discrimination thresholds vary as a function of angular size (if angle discrimination depends on angle size rather than line orientation). Since Weber’s law has been demonstrated for a wide variety of sensory discrimination tasks [see Laming (1986) for a review], it is interesting to see whether angle discrimination also follows Weber’s law, i.e. threshold increases as angle size increases. In Experiment 2, discrimination thresholds for line orientation were measured for a range of orientations. The first purpose of Experiment 2 was to verify that observers in this study do show an “oblique effect” for the orientation discrimination of lines. The second aim of Experiment 2 was to predict angle discrimination threshold based on orientation discrimination performance. Experiment 3 serves as a control, in which the orientation of the angle stimulus was randomized. This control is important because Snippe and Koenderink (1994) showed that a trial-to-trial randomization in the overall orientation of the stimulus is essential for measuring angle discrimination (and not discrimination of visual orientation) for angles made of dots.

In contrast to the previous study with dot stimuli (Snippe & Koenderink, 1994), we found that angle discrimination for lines depends primarily on angle size rather than line orientations for most angles. For a small region near the angle of 90 deg, angle discrimination threshold does show an “oblique effect” at some angles, but the magnitude of the “oblique effect” is far less than that of the line orientation discrimination. Recently, Regan *et al.* (1996) also measured angle discrimination thresholds with line stimuli, and found that angle discrimination cannot entirely be explained by line orientation discrimination. Their results are compared with ours in the General Discussion.

EXPERIMENT 1: ANGLE DISCRIMINATION

Methods

In Experiment 1, we measured angle discrimination thresholds—the observer’s ability to judge whether the angular size of an angle stimulus is greater or smaller than a reference angular size.

Two examples of angle stimuli with an angle size of 90 deg, one vertically and one obliquely oriented, are illustrated in Fig. 1. The vertically oriented stimulus

*Throughout the paper, orientations for both angle as well as the components of angle are defined as the counterclockwise angular rotation from the horizontal, unless specified otherwise. Note that when we refer to an oblique effect of angle discrimination, we mean angle components, rather than the orientation of the angle bisector.

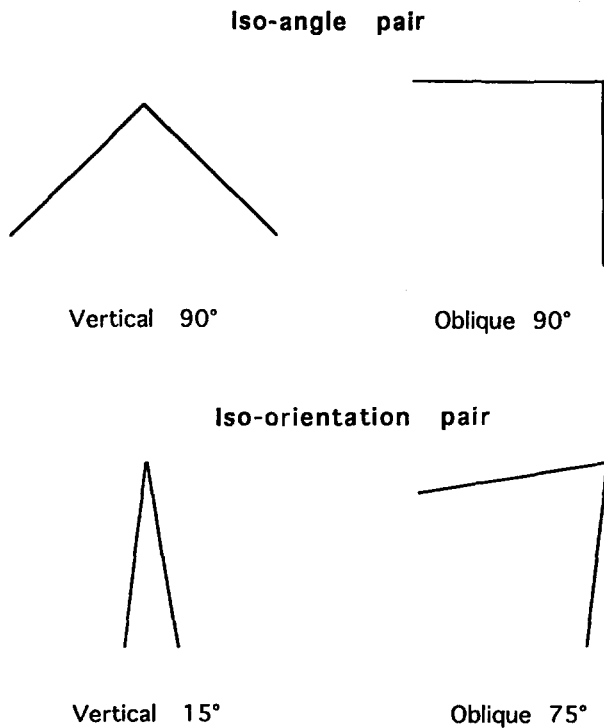


FIGURE 1. Diagrams of the stimuli used in angle discrimination. The top panel (iso-angle) shows a pair of 90 deg angles, one oriented vertically (left) and the other oriented obliquely (right). Note that these stimuli have the same angle size, but the bounding lines of the vertical angle are in oblique meridians while the bounding lines of the oblique angle are in horizontal and vertical meridians. The bottom panel (iso-orientation) shows a pair of angles with different angular sizes but the bounding lines in both angles deviate by the same angle size from the principal meridian.

looked like an inverted “V”, consisting of two bright [0.018 cd/m; see the section on calibration of Klein & Levi (1985) for details], thin, touching lines, each about 0.5 deg long. The stimuli were presented on an HP 1345A digital oscilloscope with a white P31 phosphor and a resolution of 2048 by 2048 pixels in an area of 8.5 cm vertically by 11.5 cm horizontally. The oscilloscope had a uniform dark background. The stimuli were generated by an Amiga computer through an AMIHP custom interface board. A circular aperture cut from a white square of cardboard was affixed to the screen. This circular aperture helped observers with fixation and prevented observers from using the vertical and horizontal edges of the rectangular oscilloscope as a reference. At the 3 m viewing distance, the visible part of the screen was a circle of 1.8 deg in diameter, the cardboard square subtended 5.2×5.2 deg, and 1 pixel subtended 3.8 by 2.8 sec arc. Background and stimulus luminance were measured with a Pritchard Spectra photometer. All stimuli were presented for 400 msec with an abrupt onset and offset. Timing characteristics were calibrated using a storage oscilloscope at the output of AMIHP board. All experiments were conducted with normal room illumination.

Two paid undergraduate students (QV and TN), who were naive as regards the purpose of the experiment, and one of the authors (SC) served as observers. Both naive

observers had previous experience with psychophysical experiments. All observers had or were corrected to normal vision; QV and SC had low degrees of myopia (<2.5 D), and TN is emmetropic.

Observers viewed the center of the display monocularly, while the nonviewing eye was occluded with a black patch. Head position was restrained using a chin-rest and forehead bar. Angle discrimination thresholds were determined by a signal-detection rating scale method of constant stimuli (Levi & Klein, 1983) with visual feedback. The measurement of a threshold for one reference angle (one run) consisted of at least 10 practice trials followed by 125 test trials. There was no fixation target between or during trials; the observers were instructed to fixate on the center of the circular aperture. Observers initiated trials by pressing a button on the response box. After about a half second, a test angle was presented in the center of the circular aperture for 400 msec. The test angle in each trial was randomly selected from a set of five test angles which were evenly and symmetrically disposed around the desired reference angle. After the presentation of the stimuli, the observer rated the size of the displayed angle as one of the five test angles by pressing one of five buttons. The observer had the option of initiating a new trial without making a judgment if he/she did not properly attend to the stimulus or made an eye movement. Following the observer's response, visual feedback was presented for about 250 msec. The visual feedback consisted of a short vertical line displayed on the upper portion of the screen. The horizontal location of the line indicated the relative size of the previously displayed angle. Although the observer's task was to categorize or classify the perceived angular size of the stimulus, and there was no explicit reference angle provided, it is possible for observers to establish an accurate internal reference based on feedback. In a pilot experiment, we found that angle discrimination performance in one observer (SC) was not improved by providing either an explicit spatial or a temporal reference [see also Snippe & Koenderink (1994)].

All observers received extensive practice (at least 540 trials for each angle size) at performing the angle discrimination task, prior to data collection. The pre-determined angular separations between the neighboring test angles were derived on the basis of pilot data so that the d' for the smallest increment would be about 1. The angular separations were varied, depending on which reference angle and which observer was to be tested, in the approximate range of 0.5–3.5 deg. To prevent observers from using the separation between the two line endpoints as a cue, the length of both lines was randomly varied by the same amount between 0.45 and 0.55 deg arc from trial to trial. Observers were instructed explicitly to judge the angle size rather than the end-point separation. Threshold estimates were obtained using the ROCFLEX signal detection analysis program (Klein & Levi, 1985), which makes a maximum-likelihood estimate of the d' values for each stimulus and constructs

a psychometric function relating d' to the angular separation. Thresholds were obtained by interpolating to a d' of 1 (corresponding to 84% correct) from psychometric functions where the exponent or slope of d' vs angle was held to be 1. In most cases the d' for the smallest increment was in the range of 0.8–1.2. Data sets with $d' > 1.4$ (tasks were too easy) or $d' < 0.6$ (tasks were too difficult) were discarded. The final results were the average of at least three runs weighted by the inverse standard error (Klein, 1992).

Discrimination thresholds for 12 reference angles (15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165 and 180 deg) were measured for angles presented in two orientations: one centered about vertical, the other centered 45 deg from vertical (tilted) (Fig. 1). Testing order was randomized.

Results and discussion

In this experiment, our observers had to indicate whether an angle was larger or smaller than the implicit reference angle. If angle discrimination is based on discrimination of the orientations of the two bounding lines, we would expect that angle discrimination should depend on the orientation of the bounding lines and show an oblique effect. Since orientation discrimination curves are roughly symmetrical around the two oblique (45 and 135 deg) and two principal meridians (90 and 180 deg) (Heeley & Timney, 1988; Orban *et al.*, 1984a), the curves of angle discrimination vs angle size for vertical and tilted angles should be symmetrical around the angle of 90 deg, but out of phase. This hypothetical prediction is illustrated schematically in Fig. 2, which shows a peak and a valley at the angle of 90 deg for the vertical and tilted angle respectively (Fig. 2). Alternatively, if angle discrimination depends on angle size rather than line orientations, the angle discrimination thresholds measured for vertical and tilted angles should be the same, and the two curves in Fig. 2 would overlap.

In Fig. 3, angle discrimination thresholds are shown for three observers (SC, QV, and TN) in three panels. The thresholds are plotted as a function of angle size for the vertical (solid squares) and the tilted (open circles)

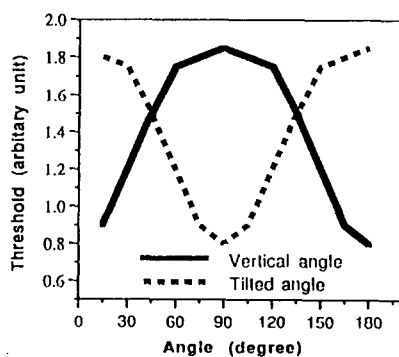


FIGURE 2. Schematic prediction of angle discrimination thresholds based on orientation discrimination thresholds. Predictions are based on data from Fig. 2 of Orban *et al.* (1984a), assuming the orientation threshold vs orientation curve is symmetrical about the oblique (45 and 135 deg) and principal (0 and 90 deg) meridians.

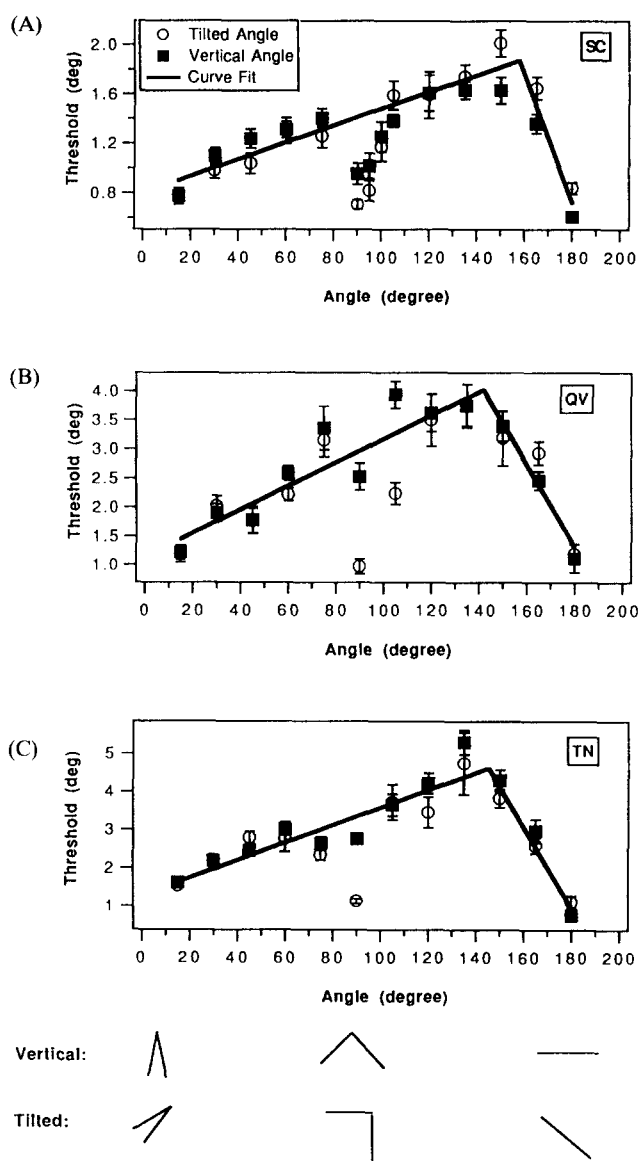


FIGURE 3. Angle discrimination thresholds obtained for vertically (■) and obliquely (○) oriented angles are plotted as a function of angle size. Data from three observers (SC, QV, and TN) are shown. The solid line shows the best fitting curves using two linear functions (see text for details). The bottom panel illustrates graphically three angle sizes at two stimulus orientations.

angles. Obviously, the overall curve pattern in Fig. 3 is quite different from that of Fig. 2. Instead of being the inverse of each other (Fig. 2), the curves for the vertical and tilted angles show very similar shapes—they both increase from 15 to about 150–160 deg (except at 90 deg), and then decrease to 180 deg. Though the shapes of the curves for the vertical angles look more similar to that in Fig. 2, they are different in two aspects. First, there is a dip at the angle of 90 deg (more evident in observers SC and QV). This dip is not expected because orientation acuity is usually worst at 45 and 135 deg meridians (Heeley & Timney, 1988; Orban *et al.*, 1984a). Second, the curves are not symmetric about the right angle (90 deg), showing higher thresholds at obtuse angles than at the acute angles. Thus, based on the overall

TABLE 1. Parameters obtained from fitting the data of Fig. 3 with linear functions [Eqn (1)]

Observer	k_1	k_2	α_0	α_1	$\alpha_2 + 180 \text{ deg}$	$k_1\alpha_1$	$k_2(\alpha_2 + 180 \text{ deg})$
SC	0.007 ± 0.001	-0.052 ± 0.008	157.9 ± 2.4	115.8	-13.7	0.78	0.712
QV	0.020 ± 0.001	-0.070 ± 0.009	142.0 ± 4.6	57.0	-18.8	1.14	1.316
TN	0.023 ± 0.001	-0.105 ± 0.007	145.3 ± 1.8	55.2	-9.2	1.26	0.966
Mean	0.0166	-0.075	148.4	76	-13.9	1.06	0.998

difference in Figs 2 and 3, we conclude that angle discrimination is not simply based on the orientation discrimination of its bounding lines.

Primacy of dependence on angle size rather than line orientation

A major concern of this experiment is whether angle discrimination depends on the orientations of the bounding lines or angle size or both. This question can be answered by examining the iso-angle (same angle size, different orientation) and iso-orientation (same orientation, different angle size) pairs (see Fig. 1). If angle discrimination depends completely on angle size, the threshold should be the same for iso-angle pairs. If angle discrimination depends completely on its line orientations, the threshold should be the same for iso-orientation pairs. The vertical and tilted angles of 15, 75, 90, 105, 165 and 180 deg are useful iso-angle pairs for the present purpose because the line orientations of these angle pairs are near the vertical or horizontal meridians in one case and near oblique meridians in the other case, and if there is any dependency on orientation, it is most likely to show in those angle pairs. By inspection of these curves (Fig. 3), it appears that the thresholds for the vertical and the tilted angles are very similar except around the angle of 90 deg. A paired *t*-test with pooled data across observers confirms this impression: the difference between the thresholds of vertical and tilted angles is significant only at 90 deg ($P < 0.001$). These results suggest that angle discrimination depends primarily on angle size, with the exception of 90 deg. A further examination of the iso-orientation pairs also supports this notion. For example, although the vertical 15 and 165 deg, tilted 75 and 105 deg (illustrated in the lower panel of Fig. 1) have similar component orientations, the thresholds are substantially lower for the vertical 15 deg angle than other angles in the iso-orientation pairs for all three observers.

Weber's law for angle?

Since angle discrimination depends primarily on angle size, we now pursue further the question of whether angle discrimination follows Weber's law—the proportionality between the discrimination threshold and the stimulus magnitude [see Laming (1986) for a review]. Weber's law is prevalently found in the spatial position domain, i.e. bisection, separation (length), and alignment discrimination task [see Burbeck (1991); Levi & Klein (1990, 1992) for reviews]. Figure 3 indicates that there may be a Weber's law for angle discrimination. If we

ignore the data near 90 deg for the moment, it is clear that the threshold increases with angle size up to 135–150 deg, which roughly follows Weber's law. Furthermore, even for the decreasing portion of the curve (150–180 deg), angle discrimination may follow Weber's law if we consider 180 deg as 0 deg and define the angle size as the complement of the angle, i.e. how many degrees the stimulus deviates from a straight line. We propose that angle discrimination generally follows Weber's law, however, Weber's law fails near 90 deg. Based on this assumption, the data are fit with two linear functions:

$$\Delta\alpha = k_1(\alpha + \alpha_1) \quad \text{if } \alpha < \alpha_0$$

$$\Delta\alpha = k_2(\alpha + \alpha_2) \quad \text{if } \alpha \geq \alpha_0$$

$$\alpha_2 = (k_1/k_2)(\alpha_0 + \alpha_1) - \alpha_0 \quad (1)$$

where $\Delta\alpha$, the dependent variable, is angle discrimination threshold; α , the independent variable, is the angle size; k_1 and k_2 are the slopes of the linear increasing and decreasing functions; α_1 and α_2 are the angle sizes at which the two linear functions intersect with the x -axis; α_0 is the angle size where the two linear functions intersect with each other. Because data at 90 deg for observers QV and TN, and at 90, 95 and 100 deg for observer SC obviously deviate from the linear function, they were not included for fitting, and because the thresholds for the vertical and the tilted angles are so close, they were fit by the same function. The data were fit separately for each observer and the estimates of the parameters, where all except α_2 were free to vary, are given in Table 1. These fits are shown as the solid lines in Fig. 3, and the fits look reasonably good (as indicated by χ^2 and by the parameter error estimates). For all three observers, slope k_2 is steeper than slope k_1 , suggesting that two different mechanisms may be involved. Note that for all observers the constant α_1 and $\alpha_2 + 180 \text{ deg}$ are not zero. Originally, Weber's law was defined as $\Delta\alpha/\alpha = k$, where k is a constant. However, there are many violations of Weber's law. Fechner (1860) found that in some cases $\Delta\alpha/(\alpha + \beta) = k$, where β is a small constant, gave a better fit to sensory discrimination data. Early studies on visual separation discrimination (Volkmann, 1863) also reported that β was not zero, and Fechner took this to indicate an irreducible error of adjustment of visual extents which he called Volkmann's Constant. This Volkmann's Constant is expressed as $k_1\alpha_1$ and $k_2(\alpha_2 + 180 \text{ deg})$ in Table 1. Thus, in terms of increase in threshold with stimulus magnitude, our data can be empirically approximately described by Weber's law despite the nonzero value of α_1 and $\alpha_2 + 180 \text{ deg}$

(though other function, e.g. power functions, may also fit our data well, see our Discussion). Note also that the strict proportionality ($\beta = 0$) in Weber's law for *position* has a meaningful consequence—scale invariance: we can increase or decrease the viewing distance without altering the observer's perception of the stimulus (Koenderink, 1977). However, the significance of strict proportionality in Weber's law for *angle* is not so clear because angle size itself is a scale invariant property.

Data at right angle (90 deg)

In general, our data seem to be fit reasonably well with two linear functions (Weber's law) and angle discrimination depends primarily on angle size rather than line orientations. However, thresholds at 90 deg (right angle) obviously deviate from Weber's law and depend more on the line orientations. For all three observers thresholds are substantially worse when the two components of the angles fall in oblique meridians (vertical right angle) than when one component falls in the vertical and the other in the horizontal meridian (tilted right angle). To see whether this oblique effect also exists for angles near 90 deg, additional data (95 and 100 deg angles both vertically and obliquely oriented) were collected for observer SC. We can see that as the angle deviates from 90 deg, the magnitude of the oblique effect is reduced [Fig. 3(A)]. This result suggests that as an exception to Weber's law for angle, there is an orientation regime near the angle of 90 deg in which threshold depends on orientation rather than angle size.

Another feature of the data near 90 deg is that the thresholds at the vertical 90 deg angle (with the bounding line oriented 45 and 135 deg obliquely) are lower than the thresholds at the vertical 75 and 105 deg for observers SC and QV. This lower threshold at angle vertical 90 deg is not expected from the meridional anisotropy in orientation discrimination: discrimination is best at 0 and 90 deg and falls monotonically to a minimum at 45 and 135 deg (e.g. Orban *et al.*, 1984a,b; Heeley & Timney, 1988). However, Regan and Price (1986) reported that orientation discrimination performance for gratings falls periodically, rather than *monotonically*, which result in the worst threshold occurring at angles other than 45 and 135 deg. The periodicity in orientation discrimination was attributed to a limited number of broad bandwidth orientation-sensitive neural elements. Similarly and more relevant to the present study, Onley and Volkman (1958) found that the precision of adjusting a line to be perpendicular to a reference line was the highest when the reference line was oriented vertically or horizontally, and the lowest precision occurred at oblique orientations adjacent to 45 and 135 deg rather than at 45 and 135 deg. This result was accounted for by a primary set of internal orientation references at 0 and 90 deg, and a secondary set of references at 45 and 135 deg. Our data verify this early observation.

In summary, our data suggest two regimes in angle discrimination. In the Weber's law regime, angle

discrimination depends primarily on angle size and is almost invariant of line orientations. In the orientation regime (near the angle of 90 deg), angle discrimination depends on line orientation and shows an oblique effect.

EXPERIMENT 2: LINE ORIENTATION DISCRIMINATION

In Experiment 1, we assumed that there were oblique effects in orientation discrimination for our observers, and showed that angle discriminations did not depend on the orientations of the bounding lines (with the exception of angles near 90 deg) and therefore concluded that there was no oblique effect for angle discrimination. However, previous studies showed that the magnitude of the oblique effect in line orientation discrimination depends on several factors. The oblique effect decreases with practice (Vogels & Orban, 1985), decreases with decreasing line length (Orban *et al.*, 1984a), and is strongly reduced in peripheral vision (Vogels *et al.*, 1984). One purpose of Experiment 2 is to demonstrate that our observers do in fact show oblique effects in line orientation discrimination with our stimuli and paradigm. The second purpose of Experiment 2 is to attempt to make quantitative predictions about angle discrimination thresholds based on orientation discrimination performance, and to compare the predicted thresholds with the measured thresholds.

Methods

The apparatus, observers, and the procedure were exactly the same as those of Experiment 1. The only difference was that the stimuli contained only one line of the angle. As a consequence of preserving the same stimulus location as in Experiment 1, the single line was not presented at the center of the circular aperture. The observers, however, were still instructed to fixate at the center of the screen to keep the retinal location of the stimulus the same as in Experiment 1. The line lengths were also jittered randomly from trial to trial as in Experiment 1. The same signal-detection rating scale method was used to determine thresholds. The line orientation on each trial was randomly selected from a set of five test orientations which were evenly and symmetrically disposed around the desired reference orientation. The observer's task was to rate the line orientation by pressing one of five buttons. Note that although this multiple response classification paradigm [see Graham (1989) for definition] has been used extensively in measuring hyperacuties such as vernier and bisection tasks (Klein & Levi, 1985; Levi & Klein, 1983), this seems to be the first time that it was applied to the orientation task. Different paradigms may affect the decision rule, which can result in different thresholds (Johnson, 1980). However, previous studies showed that a single interval recognition task and a two-interval forced choice task yielded similar orientation discrimination thresholds for vertical stimuli (Heeley & Buchanan-Smith, 1990), and orientation discrimination anisotropy

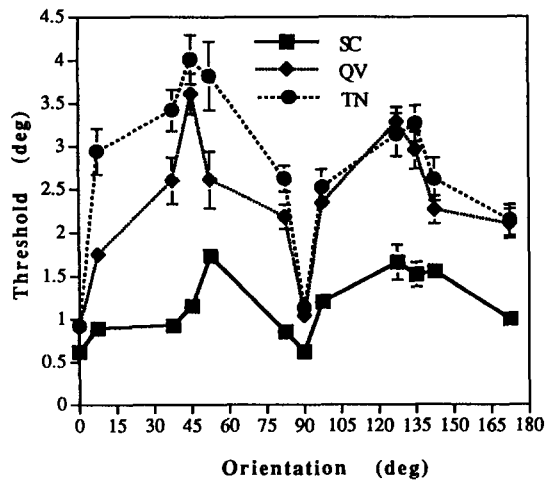


FIGURE 4. Line orientation discrimination thresholds for observers SC, QV, and TN are plotted as a function of line orientation.

cannot be simply attributed to decision factors (Vogels & Orban, 1986).

Discrimination thresholds were measured for six pairs of orientations (82.5 and 97.5 deg, 37.5 and 52.5 deg, 45 and 135 deg, 0 and 90 deg, 7.5 and 172.5 deg, 127.5 and

142.5 deg) which correspond to the bounding line orientations of the vertical and the tilted 15, 90, and 165 deg angles.

Results

As expected, all three observers demonstrated oblique effects in line orientation discrimination. Figure 4 shows the discrimination thresholds plotted as a function of orientation for three observers. All observers showed better orientation discrimination thresholds for vertical and horizontal orientations than for the oblique orientations, confirming previous studies of line orientation discrimination (Heeley & Timney, 1988; Orban *et al.*, 1984a). However, the magnitude of the oblique effect was quite different among observers. Similar variation in orientation acuity among observers was reported previously (Heeley & Timney, 1988). The thresholds for observer SC are in close agreement with the averaged thresholds of a previous study by Heeley and Timney (1988) who used a line length of 2.5 deg arc and a temporal two-alternative forced choice (2AFC) paradigm. The thresholds estimated by Orban *et al.* (1984a), however, were higher than the present study, although under one of their conditions the same line length (0.5 deg arc) was used. The higher thresholds might be due to the dimmer background and stimuli used in their study.

One of the major questions of Experiment 2 was whether angle discrimination thresholds can be predicted from orientation discrimination thresholds. To make such predictions, a simple two-stage model was constructed. [The present statistical model should be distinguished from the two-stage model proposed by Bowne (1990)]. In the first stage, the orientations of the two bounding lines are extracted in parallel, but independently of each other (as if only one line were presented). The orientation of each line might be extracted by the weighted mean of all orientation selective channels at each line location (Coltheart, 1971; Howard, 1982), but the detailed mechanism underlying line orientation extraction is not our major concern in the present study. At the second

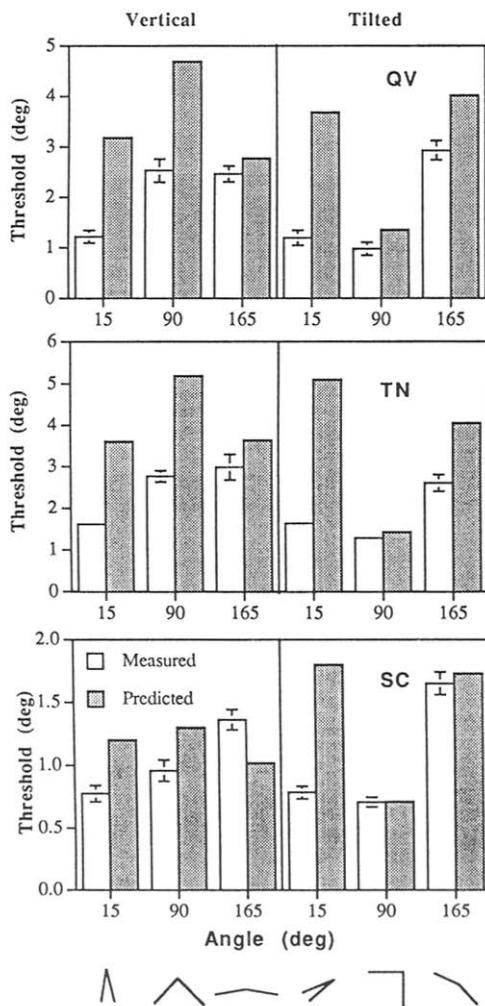


FIGURE 5. Angle discrimination thresholds (white bar) measured from Experiment 1 and angle discrimination thresholds (gray bar) predicted on the basis of line orientation discrimination (Eqn (2)). Data from three observers (SC, QV, and TN) for three angle sizes (15, 90 and 165 deg) at vertical and oblique orientations are shown.

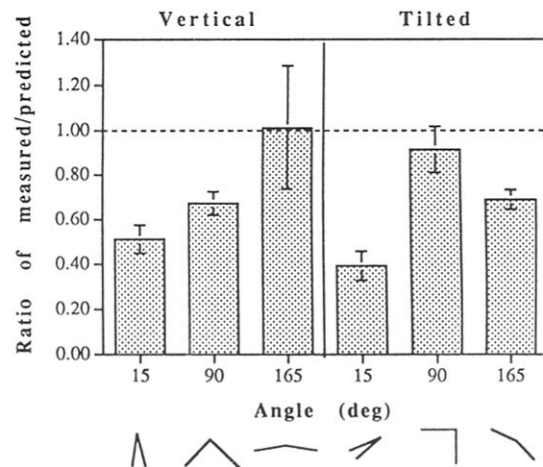


FIGURE 6. Mean ratio of the measured over the predicted thresholds for the three observers in Fig. 5. Dashed horizontal line indicates where the measured and predicted thresholds are equal.

stage, the two orientation signals are combined to compute (by subtraction) the angle. The precision of angle discrimination is assumed to be limited by the signal-noise ratio at the stage of orientation coding, and the noise that perturbs the output of the combination process is assumed to be smaller and thus not the limiting factor. Now, suppose that the noise or variances in the orientation coding are t_1^2 and t_2^2 for the two bounding lines that form the angle. t_1^2 and t_2^2 are assumed independent of each other. From basic statistics, the variance of the angle coding would be $t^2 = t_1^2 + t_2^2$. Therefore, the predicted threshold for angle discrimination is:

$$t = \sqrt{t_1^2 + t_2^2}. \quad (2)$$

The main finding of our experiment is that angle discrimination thresholds cannot be predicted from orientation discrimination thresholds, except for angles with bounding lines at or near horizontal or/and vertical meridians. Figure 5 shows the predicted and measured angle discrimination thresholds for our three observers. The white bars represent the actual measured thresholds and the gray bars represent the predicted thresholds. A summary figure showing the mean ratio of the measured/predicted threshold is shown in Fig. 6. For all observers, the measured thresholds were lower than the predicted thresholds for vertical 15 and 90 deg angles and tilted 15 and 165 deg angles. However, the measured and predicted thresholds were very similar for the vertical 165 deg angle and tilted 90 deg angle; in these cases, the orientations of the bounding lines were at or near the horizontal and/or vertical meridians. This similarity between the measured and predicted thresholds at these angles indicates that the angle size may be extracted in a manner like that described in Eqn (2), but it rules out the possibility that angles are computed by performing orientation discrimination of a single line because otherwise angle discrimination thresholds would be a factor of two higher than orientation discrimination thresholds (i.e. a 1 deg change in angle would result in 0.5 deg change in each line's orientation).

What are the predictions for stimuli of the same angular size but with different line orientations? By comparing the predicted thresholds between the vertically and obliquely oriented angles, we can see that for all observers the predicted thresholds were better when the line orientations were near the principal meridians than near oblique meridians (Fig. 5). These predicted results are *not* in agreement with the measured results, except for 90 deg angles. Note that although the predicted and measured thresholds are similar for the tilted 90 deg angle, the predicted threshold for the vertical 90 deg is substantially higher than that measured. Consequently, the magnitude of the oblique effect was much smaller for the measured thresholds than for the predicted thresholds [see also Heeley & Buchanan-Smith (1993)].

In summary, angle discrimination thresholds cannot be simply predicted from orientation discrimination in most cases. However, the thresholds for right and obtuse angles with their bounding lines at or near the horizontal and vertical meridians can be predicted reasonably accurately from orientation discrimination thresholds. Here the precision of orientation discrimination may set a limit on angle judgments.

EXPERIMENT 3: ANGLE DISCRIMINATION WITH ORIENTATION RANDOMIZATION

This experiment serves as a control. In our Experiment 1, angle stimuli were presented at a fixed orientation throughout a run. A potential problem may arise with this kind of presentation because the line orientations covary with angle, and observers may judge orientation instead of angle.* This is especially true if the visual system is more sensitive to changes in orientation than in angle. Therefore, introduction of a trial-to-trial randomization in the overall stimulus orientation may be necessary to measure thresholds for angle discrimination *per se*. Indeed, Snippe and Koenderink (1994) found that for angles made of dots, angle discrimination thresholds were significantly impaired by orientation randomization. The main purpose of this experiment was to determine the effect of orientation randomization on discrimination of angles made of lines rather than dots. Besides, the question of whether angle discrimination is susceptible to orientation jitter itself is very interesting. If angle discrimination thresholds are not affected by orientation jitter, then angle judgments must be made at a level of representation beyond the initial filter stage, where the angle size has been made explicit.

Methods

The apparatus, stimuli, and procedure were the same as in Experiment 1, except that the orientation of the angle was varied randomly up to ± 6 deg around the stimulus's nominal orientation for each trial. A complete set of measurements was conducted on observer SC. The measurements consisted of 12 angular sizes, each tested under four conditions: vertical angle with and without orientation randomization, tilted angle with and without orientation randomization. These four conditions were tested in a random order and on the same day for each angle. For observers QV and TN, thresholds were measured at a subset of the twelve angles and two orientations with orientation randomization.

Results and discussion

In contrast to the previous finding with angle stimuli made of dots (Snippe & Koenderink, 1994), we found that the introduction of random jitter of orientation did not change the shape of the angle discrimination function, though the thresholds were elevated slightly in some cases, especially for angles >90 deg. Figure 7(A) and (B) shows the results from observer SC tested with vertical and tilted angles, respectively. The open circles and the filled squares represent thresholds measured with and

*It is less likely for our observers to judge the absolute orientation rather than angle size in the Experiment 1 because the orientations of *both* bounding lines varied as angle size varied.

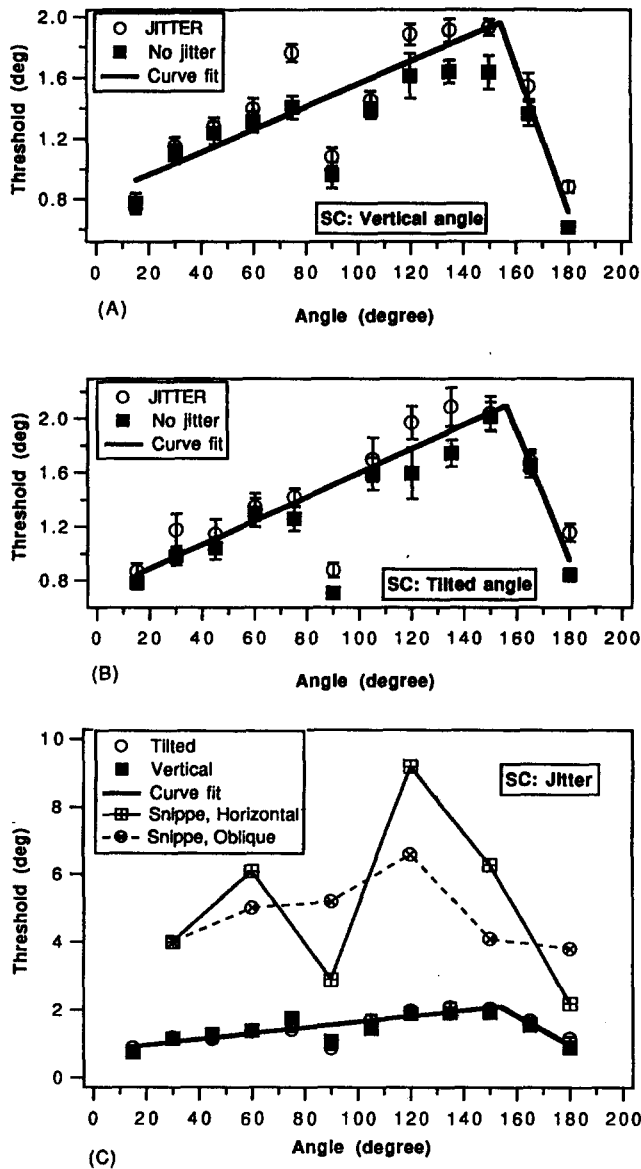


FIGURE 7. Comparison of angle discrimination thresholds measured with (○) and without (■) orientation jitter for vertical (A) and tilted (B) angles from observer SC. In (C) angle discrimination thresholds measured with orientation jitter are replotted for comparison between vertical (■) and tilted angles (○). The solid lines in the three panels are the best fit bi-linear curves. The data of one observer (HS) from the study of Snippe and Koenderink (1994) are also plotted in Fig. 7(C). Note that their definition of stimulus orientation is different from ours. For example, in their "Horizontal" condition, one arm of the angle is always near horizontal, while the other arm is rotated at an angle from this horizontal reference.

without orientation jitter. The solid line again is the curve fitting with Eqn (1). We can see that for both vertical and tilted angles, the shapes of the curves are very similar, and the thresholds are almost the same for acute angles. At several obtuse angles and near right angles, the thresholds are slightly higher with random jittering of orientation, for both vertical and tilted angles. Figure 7(C) replots the data obtained under jitter conditions for both vertical (solid squares) and tilted (open circles) angles. The curves for vertical and tilted angles are almost identical and overlapped. Comparing Fig. 7(C)

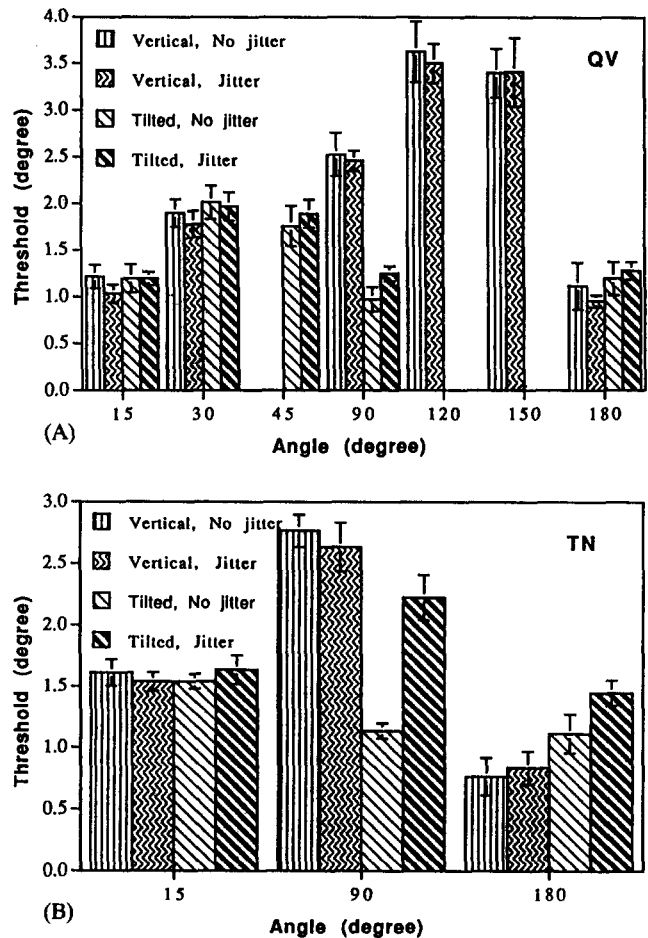


FIGURE 8. Comparison of angle discrimination thresholds measured with and without orientation jitter, at vertical or oblique orientation, for observers QV and TN.

with Fig. 2(A), which shows the data obtained in Experiment 1 without random jitter of orientation, indicates that the results are very similar. The data of one observer (HS) from Snippe and Koenderink (1994, Figs 4 and 5) are also shown in Fig. 7(C) for comparison. The thresholds obtained with line stimuli are substantially lower than those obtained with dot stimuli in Snippe and Koenderink's study. Figure 8(A) and (B) shows the results of observers QV and TN plotted as bar graphs. For observer QV the thresholds obtained with and without random jitter of orientation are almost the same. Similar results were also obtained for TN except that her threshold for the tilted 90 deg angle was substantially elevated by jitter. Thus, the results of this control experiment would not change the main conclusion made in Experiment 1.

To assure that the amount of jitter (± 6 deg) used in this experiment was sufficient to eliminate absolute orientation cues, we measured both angle discrimination for the tilted 90 deg angle and orientation discrimination at 90 deg with jitter ranging from 0 to 18 deg. Orientation discrimination was measured by turning off one arm of the angle stimuli. The result is shown in Fig. 9. In the absence of jitter, the thresholds for angle discrimination and orientation discrimination are about the same,

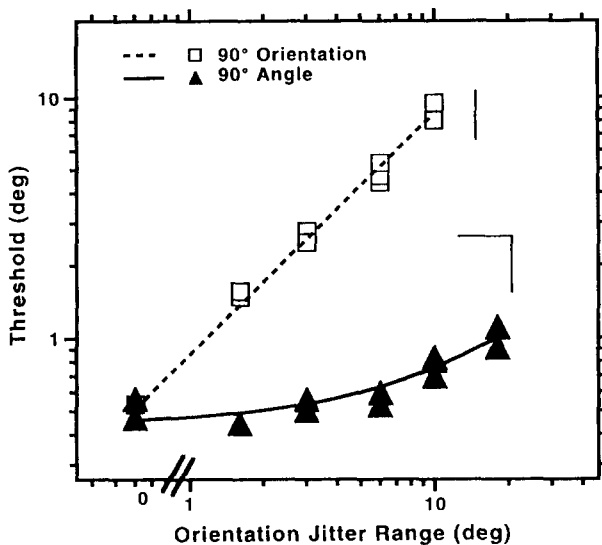


FIGURE 9. Thresholds for angle discrimination (tilted 90 deg) and orientation discrimination (90 deg) plotted as a function of jitter range in observer SC.

suggesting that angle discrimination is not based on the absolute line orientation judgment (otherwise angle discrimination threshold would be twice the orientation discrimination threshold since angle size was changed by varying both line orientations in our experiment). With a small amount of jitter (up to ± 6 deg), angle discrimination thresholds were hardly elevated, but orientation discrimination thresholds increased rapidly with the amount of orientation jitter. We can draw two conclusions from this result. First, the amount of jitter (± 6 deg) used in this experiment was sufficient to prevent observers from judging absolute line orientation (since orientation thresholds were elevated by about an order of magnitude, with almost no effect on angle thresholds); second, angle discrimination derived from the method of changing angle size by varying the orientation of both lines is a valid measurement of angle discrimination threshold *per se* [see also Regan *et al.* (1996)]. However, as orientation jitter increased beyond about 6 deg, angle discrimination thresholds were also elevated.

At least three factors may contribute to the threshold elevation observed at several angles. First, it is well documented that the perceived size of an angle varies with orientation (Wenne & Held, 1966). Horizontally and vertically oriented *acute* angles were reported to appear larger than obliquely oriented angles of the same size (Emerson *et al.*, 1975; Lennie, 1971). This bias tends to increase with line length and angular size (Hakiel, 1978). For a 45 deg angle comprising lines 0.6 deg long, the obliquely oriented angle was reported to appear 6 deg smaller than the vertically and horizontally oriented angles (Hakiel, 1978). The perceived angular size for right angles was also found to vary with orientation, with a direction and magnitude that depends on the subject (Onley & Volkman, 1958). Unfortunately, no data are available for obtuse angles. Since only a small range of orientation jitter (± 6 deg) was introduced in our experiment, the change in the perceived angle size was minimized. The second possibility is: introduction of

orientation jitter also introduces an extrinsic uncertainty effect; that is, an angle is less discriminable when an observer is uncertain about which orientation the stimulus is going to present than when he/she is certain about it. It is well established that extrinsic uncertainty about the spatial position and spatial frequency of visual stimuli can decrease an observer's *detection* performance (Cohn & Lasley, 1974; Graham, 1989). A recent study demonstrated that stimulus uncertainty can also affect suprathreshold contrast and spatial frequency *discrimination* judgments (Greenlee & Thomas, 1993). In this experiment, when a stimulus is presented in one of several possible orientations, the observer may have to either split his/her attention to monitor these possible orientations, or anticipate one possible orientation which might be wrong. Both strategies would result in the threshold being elevated. Note that the total jitter range used in our experiment was rather small (± 6 deg) compared with the orientation tuning bandwidth which varies from ± 15 to ± 30 deg, depending on spatial frequency (DeValois *et al.*, 1982; Phillips & Wilson, 1984). Thus, the "uncertainty effect" (if any) observed in this experiment may not be due to the inadequate attentional monitoring of low-level orientation selective channels. Furthermore, uncertainty should affect all angles equally, but the results indicate that orientation jitter primarily degrades discrimination only at large angles. The third possible explanation of threshold elevation is, as mentioned above, without a trial-to-trial randomization of orientation, the observer may use the orientation as a cue to judge angle. Introducing orientation jitter forces the observer to judge angle rather than orientation. One way an observer might use the orientation cue is to judge the absolute orientation of one of the lines (to which he/she is more sensitive). This strategy is more likely if one of the line orientations is fixed as a reference and angle size is varied by changing the other line's orientation. However, in our Experiment 1 the orientations of both bounding lines were changed as the angle size varied. Alternatively, an observer may judge both line orientations, and combine the orientations in the manner described by Eqn (2) described in Experiment 2. Note that this strategy should be applicable to angle discrimination even if the stimulus orientation is jittered. However, this possibility may explain the threshold elevation when the bounding lines are oriented 0 and/or 90 deg, e.g. the tilted right angle. This is because a striking feature of the threshold vs orientation curve is the precipitous fall of orientation discrimination sensitivity for orientations only a few degrees away from horizontal and vertical (Regan & Price, 1986). Thus, as a few degrees of orientation jitter (± 6 deg) are introduced in angle discrimination, on some trials the orientation of the bounding line would be oriented more obliquely than without orientation jitter. Therefore, the threshold elevation in this case may be due to the more oblique orientation rather than jitter *per se*.

In summary, the threshold elevation produced by orientation jitter may be a result of multiple factors. In

any case, the elevation is small and would not change the conclusion drawn in Experiment 1. The fact that angle discrimination thresholds are affected little by orientation jitter suggests that the angle judgments are made at a level of representation at which the angle size has been made explicit.

GENERAL DISCUSSION

Parts vs whole

In a sense, our results, that angle discrimination depends primarily on angle size rather than line orientation, are in accord with the Gestalt psychologists' famous phrase: "The whole is more than the sum of its parts." It was realized very early by Gestalt psychologists that shape (a global property) does not inhere in parts such as points and contour boundaries, but is based upon the geometrical spatial relationships among points or contour boundaries (Ehrenfels, 1890; Koffka, 1935; cited by Rock, 1986). According to this view, the spatial configuration of the two oriented lines leads to an emergent whole property—an angle, which is a simple shape. The perceptual representation of an angle is therefore different from—and not reducible to—the sum of its parts. Instead, the perception of an angle has much to do with the manner in which the two oriented lines relate to each other, i.e. how much one line is rotated away from another line. This notion is consistent with the current finding that angle discrimination depends primarily on the angular size rather than the line orientations, except for angles near 90 deg. It therefore follows that the process of combining information from the oriented two lines to form an estimate of an angle is not simple.

Our results are also compatible with a recent study on a visual search task in which a line segment of unique orientation (target) was identified among line segments of two other orientations (distractors) (Wolfe & Friedman-Hill, 1992). According to the similarity concept (Duncan & Humphreys, 1989), visual search should be easier when the orientations of distractors are similar. However, when the similarity of the distractor orientation is reduced by increasing the distractor-distractor angle, and when the target-distractor angle is smaller than the distractor-distractor angle, visual search is parallel and faster. This result can not be simply explained by the concept of orientation similarity, rather, the fast, parallel search of a different angle suggests that angle (even without apex) is a salient global feature in visual search.

Two regimes for angle and separation discrimination

One of our main findings of this study was two regimes for angle discrimination: A main angle-dependent or "Weber's law regime" in which threshold depends primarily on angle size rather than line orientation and a minor orientation-dependent regime near 90 deg in which threshold depends on line orientation. The Weber's law regime can be divided further into two sub-regimes: One in which threshold increases linearly

with angle size up to about 150 deg (the "first Weber's law regime"), another starting at 180 deg (straight line) with the threshold increasing linearly as the two bounding lines turn away from the straightness (the "second Weber's law regime") (Fig. 2). These two regimes in angle discrimination appear to be comparable to the two regimes reported in the extensively studied spatial interval discrimination task, i.e. separation and bisection. Separation discrimination is more relevant to the present study because the task is more similar: observers judge the relative position (distance) instead of angle between the two elements. With stimuli presented on an isoeccentric arc, a Weber's law regime and an eccentricity regime were revealed by measuring separation discrimination thresholds (Burbeck & Yap, 1990; Levi & Klein, 1990). When the separation of the isoeccentric test lines is small with respect to the eccentricity, thresholds are proportional to the separation of the lines (Weber's law regime); When the separation is comparable in size to the eccentricity, positional thresholds are proportional to the target eccentricity, and are essentially independent of separation (eccentricity regime). Our results in the two-dimensional orientation domain complement the findings of two regimes in the one-dimensional position domain. The orientation regime in angle discrimination and the eccentricity regime in separation discrimination are clearly different since the former depends on orientation and the latter depends on eccentricity. The Weber's law regime in angle and separation, however, share some similarities. In the first Weber's law regime, the overall separation (or area bounded by the two lines but not line ends separation, because line length is jittered) between the two lines of an angle covaries with angle size. Thus, it is possible that angle discrimination might be based upon position sensitive mechanisms. More specifically, observers may perform separation discrimination by comparing the separation at a fixed height from the apex. Since the Weber fraction for position ($\Delta x/x$) is a constant, the fixed height can be chosen arbitrarily without affecting the angle discrimination threshold. This means that angle discrimination should be independent of line length. Indeed, a previous study of angle discrimination without jittering line length reported that angle discrimination was independent of line length (Hakiel, 1978). However, our preliminary results with line length jitter indicate that angle discrimination threshold decreased as the line length increased. Therefore, with our current paradigm, it is unlikely that angle discrimination is based on separation discrimination.

Mechanisms underlying angle discrimination

Although the tasks of angle discrimination and separation discrimination are essentially different, some ideas proposed to account for position discrimination may be generalized or modified to account for angle discrimination. To reconcile the two regimes found in position judgments, Levi and Klein (1989) assumed that for any combination of separation and eccentricity, the

TABLE 2. Actual parameters used to fit data in Fig. 3 with power functions [Eqn (3)]

Observer	k_1	k_2	α_0	c
SC	0.34 ± 0.04	0.46 ± 0.17	140.5 ± 11.9	1.68 ± 0.06
QV	0.51 ± 0.08	0.41 ± 0.16	123.8 ± 13.5	3.59 ± 0.22
TN	0.52 ± 0.06	0.72 ± 0.25	141.7 ± 7.9	4.48 ± 0.21
Mean	0.456	0.53	135.3	3.25

more sensitive of the two mechanisms determines the threshold. Similarly, we propose that in angle discrimination, for any combination of angle size and orientation, the more sensitive of the two mechanisms will determine the threshold. In the eccentricity regime, the threshold was limited by the precision of locating the two elements, i.e. determined by the spatial sampling grain of the visual system (Levi & Klein, 1989, 1990), or uncalibrated neural disarray (Hess & Field, 1993; Wilson, 1991). It is assumed that once the position of the two elements is located, there is little error in the subsequent internal measurement of the separation. Likewise, in the orientation regime, the angle discrimination threshold may be limited by the precision in determining the orientations of the bounding lines. Corresponding to the factors limiting position precision, possible limiting factors in limiting orientation precision are: the sampling grain in orientation domain and the bandwidth of orientation tuning. An additional factor which may be important in the orientation domain is the internal representation of orientation.

To account for the threshold increase in the Weber's law regime for position where threshold depends on separation rather than eccentricity, several hypotheses have been proposed. For very small separations (<10 min arc), the simple single size-tuned filter model seems to fit data well (e.g. Klein & Levi, 1985; Levi & Klein, 1983; Wilson, 1986). The key notion is that there is a range of size-tuned filters in the visual system, and as the separation increases, progressively larger filters, which have shallower sloped weighting functions, will be optimally sensitive to the change in separation. As a consequence, the threshold increases with separation. This model may also predict correctly that angle discrimination threshold increases with angle size [see also Wilson & Richards (1989) for curvature discrimination]. However, since angle discrimination threshold is altered little by random jitter of orientation (especially for small angles), the differential responses generated by the size- and orientation-tuned filters would not be useful.

The second model proposes that the separation between targets is calculated by stepping from one target to the other, counting the steps as one goes (Burbeck & Yap, 1990). If each step has equal error and the steps are independent, then the Δx vs x function will have a slope of 0.5 on log-log axes (equivalent to $\Delta x \propto \sqrt{n}$). If the steps are not independent but positively correlated, the slope will be >0.5. If the steps are not independent but negatively correlated, the slope will be <0.5. If the steps

are perfectly correlated, the slope will be 1.0. To see whether this hypothesis applies to our data, the data in Fig. 2 were fitted with two power functions, one for the increasing part of the curve and one for the decreasing part of the curve in Fig. 2:

$$\Delta\alpha = c(\alpha/\alpha_0)\exp(k_1) \quad \text{if } \alpha < \alpha_0$$

$$\Delta\alpha = c\{c/(1+c)(180-\alpha)/(180-\alpha_0)\}$$

$$\exp(k_2) + c/(1+c) \quad \text{if } \alpha \geq \alpha_0 \quad (3)$$

where $\Delta\alpha$ is angle discrimination threshold, and α is angle size, c is a constant, and α_0 is the angle size where the two curves intercept, and k_1 and k_2 are the exponents of the two power functions. Note that a constant $c/(1+c)$ is added to the power function that fits the decreasing part of the curve because it is clear from inspection that the thresholds at 180 deg are not close to zero. When $\alpha = \alpha_0$, the two power functions share a common value of c . Data at 90 deg for observers QV and TN, and at 90, 95 and 100 deg for observer SC were excluded from the fitting, and the thresholds for the vertical and the tilted angles were fit using the same function. The data were fit separately for each observer and the estimates of the parameters are given in Table 2. The mean exponent (averaged over observers for both increasing and decreasing limbs) was 0.49 ± 0.05 , not significantly different from 0.5. Only one observer's exponent was significantly <0.5 (for SC the increasing function). An exponent smaller or greater than 0.5 indicates the errors in the individual steps were negatively (canceling each other) or positively correlated. Thus, our data seem to be consistent with the step-increment model. This model is attractive because it is physiologically plausible. Neurophysiological studies reveal a modular structure and columnar organization in the primary visual cortex (V1) in both cat and monkey (Hubel & Wiesel, 1962, 1974). An orientation hypercolumn contains cells sequentially arranged and tuned to the full range of orientations. The pattern of activity across the orientation columns in visual cortex has been proposed to account for the perceived orientation of visual contours (Paradiso, 1988). Here, we propose that one way to calculate angle size is to count the number of neurons between the peak activity across the hypercolumns.

The third idea was proposed by Morgan and Watt (1989) to explain their arc length discrimination data. They proposed that Weber's law for spatially separated features might result from higher level processing constraints, i.e. memory, or higher representational and internal reconstruction processes. Levi and Klein (1989) suggested that this may be true for certain tasks (e.g. arc length) which are constrained by the cognitive demands of the task rather than by sensory limitations. A possible site of such higher level neuronal processing is the inferotemporal cortex (IT), a brain structure that is thought to be essential for object vision [see Tanaka (1993) for a review]. As in V1, IT is organized in columnar modules in which cells with overlapping but slightly different selectivity cluster together. However,

the critical features that are required to activate the IT cells are more complex (e.g. triangle, circle, and square) than orientation and spatial frequency (size), which are known to be extracted and represented by cells in V1. Compared with other hyperacuity tasks (e.g. vernier alignment, bisection), the angle discrimination task is rather difficult: both line length and orientation are jittered, and there is no stable internal reference except for the angle of 180 deg and the tilted 90 deg. Furthermore, angle discrimination thresholds are immune from orientation jitter in our study, and the thresholds for our two observers QV and TN are rather high. Because of these factors we can not exclude the possibility proposed by Morgan and Watt. To explain the Weber's law for angle with this notion, one must assume that the precision of high level representation of angle size increases with angle; however, it is not clear how this explanation would lead to an exponent near 0.5.

Angle discrimination in the second Weber's law regime may share a common mechanism with curvature discrimination at low curvatures. Intuitively, discrimination at low (gentle) curvatures might be processed by units with larger receptive fields than discrimination at sharp curvatures. However, Wilson and Richards (1989) showed that discrimination thresholds at low curvatures were not affected by high-frequency bandpass-filtering, but were substantially elevated by low-frequency bandpass-filtering, suggesting high-spatial-frequency (small receptive field), orientation-selective mechanisms dominate low curvature processing. Therefore, they proposed that gentle curvatures were estimated by comparing orientations at points displaced a fixed distance along the tangent to the curve. This estimation is essentially a computation of obtuse angle. It would be interesting to see how bandpass frequency-filtering affects angle discrimination.

Comparison with other studies

Although extensive studies have been conducted on the perceived angular size (constant error, or bias) (see Howard, 1982), relatively few studies exist in the literature about the *precision* of angle perception. Hakiel (1978) measured angle discrimination thresholds for *acute* angles between 15 and 60 deg at a couple of stimulus orientations and with different line lengths. The thresholds were measured in the presence of a reference angle. A potential problem with this study is that neither the angle orientation nor the line length were randomly jittered (see our discussion on *Two regimes for angle discrimination*). For all stimulus orientations and all line lengths, Hakiel found that the angle discrimination threshold increases with the size of the angles to be compared. In agreement with the present findings, angle discrimination was independent of angle orientation. He concluded that angle discrimination thresholds are not determined simply by the combined orientation acuities for the lines comprising the angles [see also Heeley & Buchanan-Smith (1993)]. However, a more recent study (Snippe & Koenderink, 1994) using angles made of dots

showed that angle discrimination can be well understood from the meridional anisotropy for orientation discrimination of the orientation that defines the angle. Observers in their study did not even show any special sensitivity to angles of 180 deg (straightness). Also, their thresholds for angle discrimination are much higher than ours. This difference is presumably due to the different stimuli used (i.e. lines vs dots, different amount of jitter in line length and orientation). Recently, Regan *et al.* (1996) measured angle discrimination thresholds at angles between 20 and 160 deg. They employed precautions to prevent observers from judging absolute line orientation and separation between the line endpoints. They found that angle discrimination cannot entirely be explained by line orientation discrimination, which is consistent with our results. However, their results showed that the curve of angle discrimination threshold vs angle size was approximately flat between angles of 20 and 160 deg. Therefore, no Weber's law for angle discrimination was found with their experimental conditions. However, they used very large amounts of jitter for both the overall stimulus orientation (± 15 deg) and line length ($\pm 30\%$). As evident in our Fig. 9, ± 15 deg of orientation jitter not only eliminates the orientation cue, but also elevates angle discrimination thresholds. Thus, the large jitter effect (e.g. uncertainty effect) might have masked the Weber's law behavior of angle discrimination.

Implications and relation to meridional anisotropy in orientation discrimination

The rationale of this study is that if angle discrimination is based on the extraction of line orientation, angle discrimination should also depend on orientation since orientation discrimination depends on stimulus orientation. Although several suggestions have been proposed to account for the meridional anisotropy in orientation discrimination, most of them can be categorized according to two fundamental hypotheses. One hypothesis proposes that the meridional anisotropy results from the variation in the properties of orientation-selective neurons in the early stages (i.e. V1) of visual processing. The possible properties involved in causing the orientation anisotropy are orientation tuning (Andrews, 1967), neuronal sensitivity (Rose & Blakemore, 1974), and neuronal density (DeValois *et al.*, 1982; Mansfield, 1974; Mansfield & Ronner, 1978; Orban & Kennedy, 1981; Rose & Blakemore, 1974; Saarinen & Levi, 1995). An implicit assumption underlying this hypothesis is that orientation discrimination is limited by the noise in the initial filtering stage. If angle discrimination is also limited by the same noise in the initial filtering process, we would expect angle discrimination should also show an oblique effect. The fact that angle discrimination does not depend on orientation for most angle stimuli suggests that the orientation and angle discriminations may not be limited by the same noise in the early filtering stage (Heeley & Buchanan-Smith, 1994). An alternative hypothesis for meridional anisotropy is that observers can make better use of nonvisual cues, e.g. gravity and

body axis, to aid orientation discrimination when stimuli are oriented horizontally and vertically than in an oblique meridian. The main evidence supporting this hypothesis comes from the finding that the meridional anisotropy in orientation discrimination appears to be linked to gravitational rather than retinal coordinates [Buchanan-Smith & Heeley (1993); however, see also Orban *et al.* (1984a) and Chen & Levi (1995)]. According to this second hypothesis, the discrimination of orientation requires comparing the stimulus orientation with the internal representation of visual vertical and horizontal, and the neural coding of orientation therefore occurs at a higher level beyond the initial filtering operation. This notion provides a way to reconcile the different effects of orientation on angle and orientation discriminations. Orientation discrimination may be considered as a kind of angle discrimination in which one of the components of an angle is not explicitly presented, and the observer's task is to discriminate the angle formed by the stimuli and the internally represented vertical and horizontal (Buchanan-Smith & Heeley, 1993). Because angle discrimination thresholds increase with angle size at least for acute angles (Weber's law for angle), orientation discrimination thresholds should also increase as the orientation deviates from horizontal or vertical (oblique effect). Thus, both orientation and angle discrimination can be considered as relative orientation discrimination, differing only in the type of reference used. However, when an explicit reference line is presented, orientation (or parallelism) discrimination shows an oblique effect, though the effect is smaller and depends on the separation between the two lines (Andrews, 1967; Morgan, 1991). One possible explanation for this oblique effect is that the internal reference is more precise than the neural coding of the stimulus reference (Heeley & Buchanan-Smith, 1990) so when stimuli are oriented vertically or horizontally, the test orientation is compared with the internal representation of horizontal or vertical rather than the neural representation of the reference. Alternatively, the early filtering process contributes to the orientation anisotropy in the parallelism discrimination, but not in the unreferenced orientation discrimination.

Summary and conclusion

The present study shows that while there is strong meridional anisotropy in line orientation discrimination, the discrimination of an angle consisting of two oriented lines depends primarily on angle size rather than line orientation. Following the two regimes found for separation discrimination, we suggest that two regimes also exist in angle discrimination: an angle-dependent, or Weber's law regime, and an orientation-dependent regime. In the orientation regime, the angle discrimination threshold may be limited by the precision in determining the orientations of the bounding lines. When our data in the angle-dependent regime were fitted with a power function, the exponents were close to 0.5, consistent with the use of a step-increment approach to calculate angle. Also, we showed that angle discrimi-

nation thresholds are quite robust to small amounts of orientation jitter, suggesting that angle judgments are made at a level beyond the filter representation at which the angle size has been made explicit.

REFERENCES

- Andrews, D. P. (1967). Perception of contour orientation in the central fovea. Part I: Short lines. *Vision Research*, 7, 975–997.
- Appelle, S. (1972). Perception and discrimination as a function of stimulus orientation: The "oblique effect" in man and animals. *Psychological Bulletin*, 78, 266–278.
- Biederman, I. (1987). Recognition-by-components: A theory of human image understanding. *Psychological Review*, 94, 115–147.
- Bowne, S. F. (1990). Contrast discrimination cannot explain spatial frequency, orientation or temporal frequency discrimination. *Vision Research*, 30, 449–461.
- Buchanan-Smith, H. M. & Heeley, D. W. (1993). Anisotropic axes in orientation perception are not retinotopically mapped. *Perception*, 22, 1389–1402.
- Burbeck, C. A. (1991). Encoding spatial relations. In Watt, R. J. (Ed.) *Vision and visual dysfunction*, Vol. 14, *Pattern recognition by man and machine* (pp. 8–18). Boca Raton: CRC Press.
- Burbeck, C. A. & Yap, Y. L. (1990). Two mechanisms for localization? Evidence for separation-dependent and separation independent processing of position information. *Vision Research*, 30, 739–750.
- Chen, S. & Levi, D. M. (1995). Meridional anisotropy in the discrimination of parallel and perpendicular lines—effect of body tilt. *Investigative Ophthalmology & Visual Science* 36(suppl), S633.
- Cohn, T. & Lasley, E. D. J. (1974). Detectability of a luminance increment: Effect of spatial uncertainty. *Journal of the Optical Society of America*, 64, 1715–1719.
- Coltheart, M. (1971). Visual features analyzers and aftereffects of tilt and curvature. *Psychological Review*, 78, 114–121.
- DeValois, R. L., Yund, E. W. & Hepler, N. (1982). The orientation and direction selectivity of cells in macaque visual cortex. *Vision Research*, 22, 531.
- Duncan, J. & Humphreys, G. W. (1989). Visual search and stimulus similarity. *Psychological Review*, 96, 433–458.
- Ehrenfels, C. V. (1890). Ueber Gestalt qualitäten. *Vierteljahrsschrift für Wissenschaftliche Philosophie*, 14, 249.
- Emerson, P., Wenderoth, P., Curthoys, I. & Edmonds, I. (1975). Measuring perceived orientation. *Vision Research*, 15, 1031–1033.
- Fechner, G. T. (1860). *Elemente der Psychophysik*. 2 Vols. Leipzig: Breitkopf and Hartel. (Vol. 1 trans. by Adler, H. E.); *Elements of psychophysics*. New York: Holt, Rinehart and Winston, 1966.
- Foster, D. H. (1982). Analysis of discrete internal representations of visual stimuli. In Beck, J. (Ed.), *Organization and representation in perception*. Hillsdale, N. J.: Lawrence Erlbaum Associates.
- Graham, N. (1989). *Visual pattern analyzers*. New York: Oxford University Press.
- Greenlee, M. W. & Thomas, J. P. (1993). Simultaneous discrimination of the spatial frequency and contrast of periodic stimuli. *Journal of the Optical Society of America A*, 10, 395–404.
- Hakiel, S. R. (1978). Variable and constant errors in perceived angle size. Unpublished doctoral dissertation. Keele, Staffordshire: University of Keele.
- Heeley, D. W. & Buchanan-Smith, H. M. (1990). Recognition of stimulus orientation. *Vision Research*, 30, 1429–1437.
- Heeley, D. W. & Buchanan-Smith, H. M. (1992). Orientation acuity estimated with simultaneous and successive procedures. *Spatial Vision*, 6, 1–10.
- Heeley, D. W. & Buchanan-Smith, H. M. (1993). On the relationship between geometrical organisation and local orientation: Discrimination of angles. *Perception*, 22 Supplement, 35.
- Heeley, D. W. & Buchanan-Smith, H. M. (1994). Evidence for separate, task dependent noise processes in orientation and size perception. *Vision Research*, 34, 2059–2069.
- Heeley, D. W. & Timney, B. (1988). Meridional anisotropies of

- orientation discrimination for sine wave gratings. *Vision Research*, 28, 337–344.
- Hess, R. F. & Field, D. J. (1993). Is the increased spatial uncertainty in the normal periphery due to spatial undersampling or uncalibrated disarray? *Vision Research*, 33, 2663–2670.
- Howard, I. (1982). *Human visual orientation*. New York: Wiley.
- Hubel, D. H. & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *Journal of Physiology (London)*, 160, 106–154.
- Hubel, D. H. & Wiesel, T. N. (1974). Sequence regularity and geometry of orientation columns in the monkey striate cortex. *Journal of Comparative Neurology*, 158, 295–306.
- Johnson, K. O. (1980). Sensory discrimination: Decision process. *Journal of Neurophysiology*, 43, 1771–1792.
- Klein, S. A. (1992). An Excel macro for transformed and weighted averaging. *Behavior Research Methods, Instruments, and Computers*, 24, 90–96.
- Klein, S. A. & Levi, D. M. (1985). Hyperacuity thresholds of 1 sec: Theoretical predictions and empirical validation. *Journal of the Optical Society of America A*, 1170–1190.
- Koenderink, J. J. (1977). Current models of contrast processing. In Spekreijse, H. & Tweel, L. H. (Eds), *Spatial contrast, report of a workshop*. Amsterdam: North Holland.
- Koffka, K. (1935). *Principles of Gestalt psychology*. New York: Harcourt Brace Jovanovich.
- Laming, D. (1986). *Sensory analysis*. London: Academic Press.
- Lennie, P. (1971). Distortions of perceived orientation. *Nature*, 233, 155–156.
- Levi, D. M. & Klein, S. A. (1983). Spatial localization in normal and amblyopic vision. *Vision Research*, 23, 1005–1017.
- Levi, D. M. & Klein, S. A. (1989). Both separation and eccentricity can limit precise position judgements: A reply to Morgan and Watt. *Vision Research*, 29, 1463–1469.
- Levi, D. M. & Klein, S. A. (1990). The role of separation and eccentricity in encoding position. *Vision Research*, 30, 557–585.
- Levi, D. M. & Klein, S. A. (1992). "Weber's Law" for position: The role of spatial frequency and contrast. *Vision Research*, 32, 2235–2250.
- Mansfield, R. J. W. (1974). Neural basis of orientation perception in primate vision. *Science*, 186, 1133–1135.
- Mansfield, R. J. W. & Ronner, S. F. (1978). Orientation anisotropy in monkey visual cortex. *Brain Research*, 149, 229–234.
- Morgan, M. J. (1991). Hyperacuity. In Regan, D. M. (Ed.), *Vision and visual dysfunction, Vol. 10: Spatial vision* (pp. 87–113). Boca Raton: CRC Press.
- Morgan, M. J. & Watt, R. J. (1989). The Weber relation for position is not an artefact of eccentricity. *Vision Research*, 29, 1457–1462.
- Onley, J. W. & Volkman, J. (1958). The visual perception of perpendicularity. *American Journal of Psychology*, 71, 504–516.
- Orban, G. A. & Kennedy, H. (1981). The influence of eccentricity on receptive field types and orientation selectivity in areas 17 and 18 of the cat. *Brain Research*, 203.
- Orban, G. A., Vandenbussche, E. & Vogels, R. (1984a). Human orientation discrimination tested with long stimuli. *Vision Research*, 24, 121–128.
- Orban, G. A., Vandenbussche, E. & Vogels, R. (1984b). Meridional variations and other properties suggesting that acuity and orientation discrimination rely on different neuronal mechanisms. *Ophthalmic Physiology & Optics*, 4, 89.
- Palmer, S. E. (1977). Hierarchical structure in perceptual representation. *Cognitive Psychology*, 9, 441–474.
- Paradiso, M. A. (1988). A theory for the use of visual orientation information which exploits the columnar structure of striate cortex. *Biological Cybernetics*, 58, 35–49.
- Phillips, G. C. & Wilson, H. R. (1984). Orientation bandwidths of spatial mechanisms measured with masking. *Journal of the Optical Society of America*, A2, 226–232.
- Regan, D., Gray, R. & Hamstra, S. J. (1996). Evidence for a neural mechanism that encodes angles. *Vision Research*, 36, 323–330.
- Regan, D. & Price, P. (1986). Periodicity in orientation discrimination and the unconfounding of visual information. *Vision Research*, 26, 1299–1302.
- Rock, I. (1986). The description & analysis of object and event perception. In Boff, K. T. (Ed.), *Handbook of perception and human performance, Vol. II: Cognitive process and performance*. New York: John Wiley and Sons.
- Rose, D. & Blakemore, C. (1974). An analysis of orientation selectivity in the cat's visual cortex. *Experimental Brain Research*, 20, 1–17.
- Saarienen, J. & Levi, D. M. (1995). Orientation anisotropy in vernier acuity. *Vision Research*, 35, 2449–2461.
- Snippe, H. P. & Koenderink, J. J. (1994). Discrimination of geometric angle in the fronto-parallel plane. *Spatial Vision*, 8, 309–328.
- Tanaka, K. (1993). Neuronal mechanisms of object recognition. *Science*, 262, 685–688.
- Treisman, A. (1986). Properties, parts, and objects. In Boff, K. T. (Ed.), *Handbook of perception and human performance, Vol. II: Cognitive process and performance*. New York: John Wiley and Sons.
- Ullman, S. (1989). Aligning pictorial descriptions: An approach to object recognition. *Cognition*, 32, 193–254.
- Uttal, W. R. (1981). *A taxonomy of visual processes*. Hillsdale, NJ: Lawrence Erlbaum Associates.
- Vogels, R. & Orban, G. A. (1985). The effect of practice on the oblique effect in line orientation judgments. *Vision Research*, 25, 1679–1687.
- Vogels, R. & Orban, G. A. (1986). Decision factors affecting line orientation judgements in the method of single stimuli. *Perception and Psychology*, 40, 74–84.
- Vogels, R., Vandenbussche, E. & Orban, G. A. (1984). The influence of eccentricity on orientation discrimination in normal and amblyopic vision. *Perception*, 13, A23.
- Volkman, A. W. (1863). *Physiologische Untersuchungen im Gebiete der Optik*. Leipzig: Breitkopf and Hartel.
- Wenne, P. & Held, R. (1966). Changes in perceived size of angle as a function of orientation in the frontal plane. *Journal of Experimental Psychology*, 71, 55–59.
- Wilson, H. R. (1986). Response of spatial mechanisms can explain hyperacuity. *Vision Research*, 26, 453–469.
- Wilson, H. R. (1991). Model of peripheral and amblyopic hyperacuity. *Vision Research*, 31, 967–982.
- Wilson, H. R. & Richards, W. A. (1989). Mechanisms of contour curvature discrimination. *Journal of the Optical Society of America*, A6, 106–114.
- Wolfe, J. M. & Friedman-Hill, S. R. (1992). Visual search for oriented lines: The role of angular relations between targets and distractors. *Spatial Vision*, 6, 199–207.

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